

Improving Organisational Interoperability for e-Governance: A Semantic Model

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Interoperability between government agencies is critical for effective e-governance, enabling seamless communication, efficient service delivery and better use of resources. Despite Estonia's leadership in digital governance, challenges such as wasted resources and inconsistent technology governance practices persist, limiting scalability and value creation for society. To address these challenges, Estonia's three ministries have developed a strategic framework for technology governance in 2023, tailored to the specific needs of the post-COVID era, while also responding to the global and local challenges of the changing world. This paper focuses on and elaborates the second output of the Estonian Framework: Semantic & Organisational Interoperability Model - a fundamental cornerstone and success prerequisite for collaborative value creation for society and improved efficiency.

Keywords—organisational interoperability, e-governance, collaboration improvement, Estonian best practice, semantic model.

I. INTRODUCTION

The purpose of this paper is to address the challenges of interoperability between government agencies and propose practical solutions through a unified semantic model and a portfolio management framework. This introduction sets the stage by outlining the objectives, significance, and structure of the study.

A. Objectives

This study aims to:

- Establish a common terminology framework for managed objects, roles and responsibilities across agencies to promote mutual understanding.
- Analyse the current division of roles and responsibilities between business and IT to identify gaps and overlaps.
- Define the minimum viable roles and responsibilities required to ensure sustainable and scalable portfolio management practices.
- Propose a unified semantic model and portfolio management framework to improve organisational interoperability.

B. Significance

Interoperability is critical to achieving the broader goals of digital transformation, innovation and resource efficiency. The study provides actionable insights not only for Estonia, but also for other governments around the world seeking to advance e-governance systems.

C. Structure of the paper

The paper is structured as follows:

- **Background:** Discusses the global and local drivers for interoperability, including the EU Single Digital Gateway Regulation and Estonia's Digital Agenda 2030.
- **Problem Statement:** Identifies the current challenges in achieving interoperability.
- **Methods and research framework:** Describes the research design and activities, data collection methods and analytical frameworks used as well as the limitations.
- **Results:** Presents key findings, including identified gaps and proposed solutions.
- **Conclusion:** Summarises the findings, offers recommendations and suggests future research directions.

II. BACKGROUND

Interoperability is fundamental to e-governance, facilitating data sharing, coordination and integrated service delivery between government agencies. Globally, the need for collaboration is increasingly driven by frameworks such as the European Union's Single Digital Gateway Regulation [1]. This regulation mandates seamless access to key government services across EU member states, emphasising the importance of cross-border interoperability. It aims to simplify processes for citizens and businesses, particularly those related to 'life events' such as starting a business, moving to another country or accessing healthcare services. These life events require integrated workflows across multiple government agencies to ensure timely and effective service delivery.

At the local level, Estonia's Digital Agenda 2030 [2] emphasises the need for interoperability to achieve its vision of a user-centric e-state. The strategy prioritises life event-based services, which are designed to address significant moments in citizens' lives by combining data and processes from different agencies. For example, the birth of a child triggers interconnected workflows between healthcare, population registers and social security services. Effective collaboration is critical to realising this vision, as fragmented systems hinder the delivery of such comprehensive services.

Both globally and locally, the increasing complexity of citizen and business needs, coupled with rising expectations for seamless digital experiences, are forcing agencies to collaborate. This collaboration not only ensures compliance with legal mandates, but also aligns with the broader goals of efficiency, transparency and user satisfaction in e-governance. Locally, Estonia's Digital Agenda 2030 [2] reinforces the importance of inter-agency collaboration as a cornerstone of its digital transformation initiatives. Estonia's leadership in e-governance, characterised by services such as e-residency and

i-voting, underlines the critical role of interoperability. However, the implementation of life-event-based services, such as registering a newborn or managing unemployment benefits, has highlighted the challenges of aligning agency-specific processes, data management and IT systems. Without robust collaboration mechanisms, the promise of user-centred public services remains unrealised.

The convergence of global regulations and local strategic priorities has created an urgent need for innovative interoperability solutions. Achieving this requires not only technical compatibility, but also semantic alignment and organisational coherence. The European Interoperability Framework [3] also recommends that integrated public service governance should, as a minimum, include the definition of organisational structures, roles and responsibilities, and the decision-making process for the stakeholders involved. These factors are critical to the delivery of integrated services that address the complex, multi-agency processes inherent in life-event scenarios. By overcoming these challenges, governments can improve the citizen experience, streamline government operations and maximise resource efficiency.

A. Problem Statement

According to the recent study by three Estonian ministries in 2023, the problems are as follows [4].

Despite advances in digital government, agencies often operate in silos, leading to inefficiency and ineffectiveness. A critical challenge lies in misunderstanding the semantics of managed objects such as services, processes, products, resources and assets, and their interrelationships. This semantic confusion leads to inconsistent interpretations and implementations, hindering effective collaboration.

Another key issue is the lack of clarity in defining roles and responsibilities at different levels of government - ministries, agencies and development projects. This ambiguity leads to overlapping roles, gaps in accountability and inefficient use of resources. For example, agencies may appoint service owners to collaborate on the development of life-support services, but as long as responsibilities are interpreted differently, project implementation may fail.

Furthermore, the lack of roles and responsibilities for portfolio management exacerbates these problems. Effective portfolio management of managed assets, including development needs, is critical to ensuring coherence and avoiding duplication. Without it, siloed development efforts persist, leading to duplication of resources, increased waste and unnecessary complexity at the technology layer. This lack of oversight also increases risks, particularly in maintaining interoperable systems and managing technology dependencies.

Together, these challenges impede the scalability, sustainability and efficiency of achieving e-government strategic goals, making it imperative to address semantic misunderstandings, clarify roles and institutionalise portfolio management practices.

B. Research Questions

To address the challenges, the research questions are defined as follows:

- Given the diversity of development practices across organisations, what could be a common space of

terminology (managed objects, roles and responsibilities) to map current practices in a comparable way and create mutual understanding between parties?

- What is the current division of roles and responsibilities between business and IT?
- What is the list of roles and responsibilities that need to be covered as a minimum by government to ensure sustainable implementation of the model?

III. METHODS AND RESEARCH FRAMEWORK

A mixed-methods approach was adopted, combining qualitative case studies, expert interviews, and a comprehensive desktop research. This design ensures a holistic understanding of the challenges and potential solutions. The following activities were planned as shown in the table below.

TABLE I. RESEARCH ACTIVITIES AND OBJECTIVES.

Activity	Objectives
Consultation with key stakeholders	Discover the initial list of technology governance and management practices as input for the CAWI survey design and desktop research. Discover the potential list of successful ICT projects as candidates for the case studies.
Desktop research	Analyse the definitional similarities and differences of the glossaries in the relevant technology governance and management frameworks and standards in order to propose alternative semantic models for the managed objects, roles and responsibilities. Analyse the similarities and differences in the formal practices of technology governance and management within the domains of three ministries.
Case studies	Discover the success criteria for successful ICT developments.
CAWI survey	Discover the practical realities experienced by ICT governance stakeholders across government, involving both ICT and business stakeholders.
Analysis of the alternative operating models	Propose alternative models for further discussion with business and ICT stakeholders.
Selection of the final model	A well-reasoned, validated and accepted by the stakeholders' decision on the best choice of model for the Estonian context.

The strategic framework, including the Semantic & Organisational Interoperability Model, was built on the following state-of-the-art practices and standards, which represent modern technology management principles with a focus on co-creating value for end users:

- ITIL® 4 (service value system and the four-dimensional model) [4];
- BIZBOK® version 11.0 (capabilities, business architecture management maturity model, public sector reference model) [5];
- Business process management lifecycle [6];
- ISO/IEC 38506:2020 model of strategic governance and management of technology [7];

- Multinational practices of information and document management [8].

A. Limitations

Although the methodology was robust, certain limitations must be acknowledged. The reliance on professional experience and desktop research introduces an element of subjectivity, even when supported by external evidence. In addition, the CAWI survey, while highly representative of ICT management domains, captured a relatively small number of responses. This may limit the generalisability of the findings to the wider public sector.

However, as Flyvbjerg [9] argues, the depth and specificity of case-based research often compensates for limitations in sample size, especially when addressing complex, contextual issues.

IV. RESULTS

The research results were rich and provided good and valid input for mapping today's operating models of ICT developments, as well as for the development of a unified reference model, including the Semantic & Organisational Interoperability Model [10] [11] [12].

The case study interviews provided rich input on the conditions for the success criteria of ICT developments in particular. 16 case studies were conducted, involving both ICT and business stakeholders, covering cases from Estonia and abroad, from the private and public sectors. 80% of all success criteria overlapped in all cases, which gives strong validity to the proposed reference operating model – the fourth outcome of the study [11].

The relevant success criteria in the context of the current paper are presented in the table below.

TABLE II. THE RELEVANT SUCCESS CRITERIA (2 OUT OF 10)

Nr.	Success criterion	Description
3	Clarity of management system, roles and responsibilities	Each participant in the value stream understands the responsibility that comes with their role and updates their skills and competences accordingly – through all levels of management.
5	Top level managers' contribution	The involvement, participation and leadership of senior managers is critical as it supports a change in the organisation, gives mandates for the changes and measures the performance.

A. CAWI survey findings

1) Respondent demographics

The survey was completed by 48 respondents across government institutions. 30 respondents from 9 different government domains represented ICT, 18 respondents from 11 different government domains represented business.

2) Use of management practices across domains

The analysis of management practices across the different domains reveals clear trends, driven by the priorities and capabilities of organisations. Business process management (BPM) is the most widely adopted practice, with 85% of respondents reporting its use in their organisations. This reflects a strong focus on improving operational efficiency and streamlining workflows, a priority that cuts across domain boundaries. The high adoption rate underscores the universal

importance of structured process management in achieving organisational goals.

Product management follows closely behind, with 78% of respondents reporting its use. This practice is particularly prevalent in areas such as the Ministry of Justice and the Ministry of Climate, where the delivery of effective products or services to end users is a core function. The widespread use of product management underlines the importance of ensuring that offerings meet user needs and align with organisational strategies.

In contrast, service management has a lower adoption rate of 52%. While it is integral in service-oriented domains such as the Ministry of Social Affairs, its inconsistent application in other domains suggests that not all organisations prioritise or emphasise direct service delivery as part of their strategic objectives. This variation may reflect differences in operational focus, with some domains focusing more on internal processes rather than outward-facing services.

Business architecture management, with a reported usage rate of 45%, shows a more selective pattern of adoption. This practice, which involves the strategic alignment of organisational capabilities and structures, tends to be used in larger domains or those with complex mandates, such as the Ministry of Economic Affairs and Communications. The relatively low usage suggests that not all domains currently prioritise such broad strategic oversight.

Portfolio management emerges as the least used practice, with only 30% of respondents reporting its use. This suggests that structured approaches to balancing and monitoring projects, processes and services are still in their infancy in many organisations. However, its use is more evident in project-driven areas, such as the Ministry of Interior, where it supports the alignment of projects with strategic objectives.

3) Use of frameworks and principles for ICT governance and management

The data highlights a strong reliance on Agile and related frameworks such as Scrum and DevOps, which are particularly well suited to modern, iterative development practices. Meanwhile, the lower adoption of specialised frameworks such as COBIT [13] or TOGAF [14] suggests that not all organisations are yet fully embracing strategic IT governance or architecture practices.

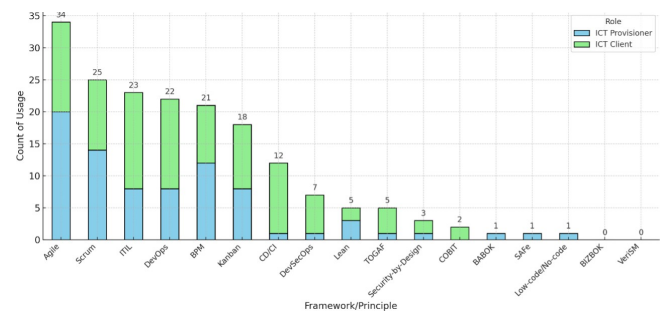


Fig. 1. Use of frameworks/principles in ICT governance, management, development and maintenance.

While the diversity of management practices highlights organisational efforts to optimise operations and achieve strategic goals, it also points to a potential gap in role clarity. The lack of a unified, standardised approach can create challenges in accountability, coordination and communication across domains. To address this, organisations can benefit

from clearly articulating the purpose and ownership of each framework and aligning them with well-defined roles at each organisational level - governance, management and operations.

4) Division of the management responsibilities

The figures below shows that while the business side takes the lead in strategy, funding, user-oriented tasks and evaluation, IT takes responsibility for implementation, and collaboration is key in areas that require alignment between business goals and technical execution.

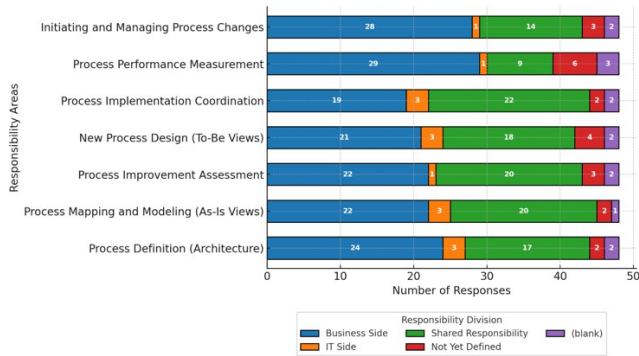


Fig. 2. Division of responsibilities for process management.

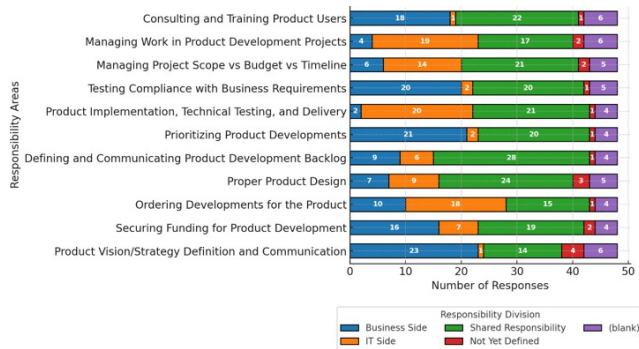


Fig. 3. Division of responsibilities for product management.

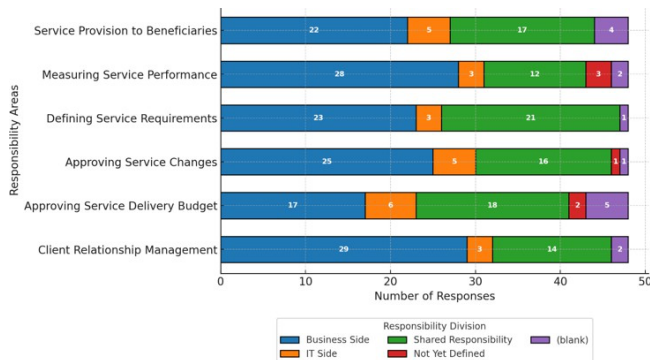


Fig. 4. Division of responsibilities for service management.

The following figure shows that IT dominates technical areas such as technical and application architecture management, while the business side leads in business-oriented tasks such as process and business architecture management. Shared responsibility is most evident in data and information architecture management, reflecting the interrelated nature of these areas. However, gaps in role clarity, particularly in service and business architecture management, suggest opportunities for improved formalisation and alignment.

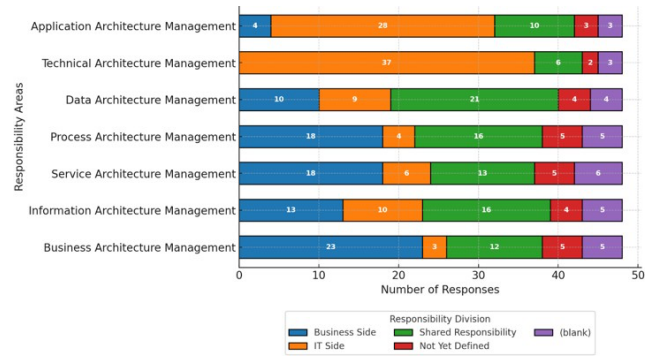


Fig. 5. Division of responsibilities for business architecture management.

The following figure shows that shared responsibility is the most significant trend across portfolio management tasks, particularly in Project Portfolio Management and Service Portfolio Management, where collaboration is essential. Business takes the lead in process portfolios, reflecting its strategic oversight, while IT takes a stronger role in product portfolios. Relatively large gaps in clarity, particularly in process and product portfolios, suggest opportunities to further formalise roles and ensure greater accountability.

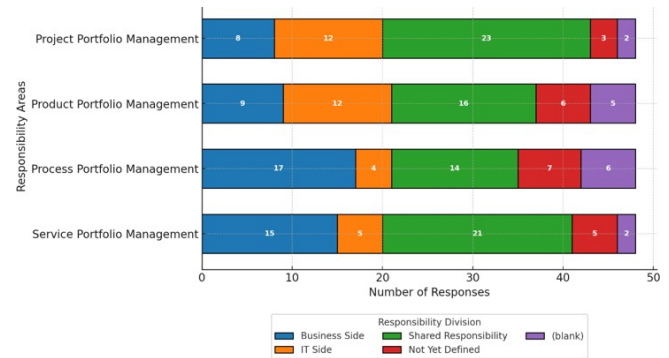


Fig. 6. Division of responsibilities for portfolio management.

5) CAWI survey summary

Of the total respondents, 32 individuals (approximately 67%) answered "no" to the use of certain management practices (section 2 of the questionnaire), but still assigned specific responsibilities under the same categories (either business side, IT side or shared responsibility in sections 3 to 7 of the questionnaire). This contradiction suggests a potential mismatch between the reported adoption of practices and the actual distribution of roles within organisations. The authors conclude that although managed objects are not clearly defined in organisations' management practices, the responsibilities for managing these objects are still being met and are necessary.

The survey results highlight a clear need to standardise terminology across management silos, as inconsistent or unclear language appears to affect understanding and role clarity. This need becomes particularly clear when we consider the terminology clarity results alongside other survey findings. When considered alongside broader survey findings such as fragmented management practices, overlapping responsibilities and shared ownership across business and IT domains, it is clear that inconsistent terminology exacerbates these challenges. Without standardisation, silos can develop their own definitions of roles, processes and priorities, leading to misalignment and inefficiency.

V. PROPOSED SOLUTIONS

A. Semantic model for the managed objects

Desktop research on the legal aspects of public service delivery and ICT governance - shows that the business architecture and its components, as well as the managed objects, are not a free choice of the ministries and agencies, but an obligation resulting from the regulations [11]. The responsibilities at different levels of government are also defined, including the responsibilities for public service portfolio management at the ministerial level [15].

The managed objects defined by the regulations are: value chain, value stream, value, service, business process, product, service portfolio, process portfolio, product portfolio.

The proposed semantic model was synthesised from different ICT governance and management frameworks and translated into the context of the Estonian public sector. The comprehensibility, usability and acceptance of the terminology was validated with all ICT and business stakeholders and ICT management representatives from 3 ministries (including the CIO of Estonia) and their sub-institutions.

The proposed terminology was based on the following conceptual model of the customised service value stream with its managed objects and their relationships. [11] [12]

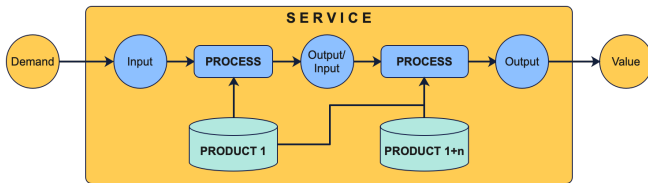


Fig. 7. Conceptual model of service value stream managed objects and their relationships.

The following table consists of the managed object definitions according to the proposed semantic model.

TABLE III. DEFINITIONS OF THE MANAGED OBJECTS.

Managed object	Definition
Service	A service is the result of the co-creation of processes and products, the operation of which delivers the desired value from the business customer (the beneficiary in terms of ICT management) to the user (the beneficiary in terms of the outcome of the provision of the service, including the user of the product). The service only exists at the time of its delivery.
Product	A product is a configuration of resources used during process activities to create value for beneficiaries. The product exists continuously, regardless of its use or non-use.
Process	A process is a set of related or interacting activities that transform inputs into outputs. A process takes one or more defined inputs and transforms them into defined outputs. Processes define the sequence of actions and their interdependencies.
Resource	A resource is a person or other physical (computer, fridge, item, object) or virtual (software, database, data) entity needed to perform an activity or achieve a goal. The resources used by the organisation may be owned by the organisation (own employees, servers) or used under contract with the resource owner (outsourced programmers, data centre).

B. Minimum Viable Role Model

Minimum Viable Role Model (MVR) - has been formulated as universal principles of essential roles and responsibilities and their division between business and IT and can therefore be applied at all levels of management - institutional, ministerial or governmental. These essential roles and responsibilities must be in place in organisations to ensure the sustainability of successful business service developments.

The table below shows the key role definitions for the managed objects.

TABLE IV. IMPORTANT ROLES RELATED TO THE MANAGED OBJECTS.

Role	Responsibilities
Service Owner	The role responsible for the delivery of a service to a user, including demand and value management, service design, process and product assurance, customer relationship management.
Product Manager	A role with full responsibility for the success of the product. The role of the product manager includes the following responsibilities - product strategy, product definition (from the development roadmap to the definition of business requirements), product realisation and product marketing. Product Managers are involved in the deployment process to ensure that the product developed meets the needs of the market. The Product Manager is market-focused and is responsible for the longer-term activities.
Product Owner	Overlaps with the Product Manager role, but does not cover all the responsibilities of a Product Manager. The Product Owner's focus is internal to the organisation - setting detailed requirements for the development team and working closely with them throughout the development process. The Product Owner represents the product to the development team and should therefore be aware of the market and the needs of the stakeholders. The Product Owner is responsible for short-term activities such as understanding the needs of service and process owners, managing the portfolio of development needs, prioritising development, creating the development backlog and managing the developers.
Process Owner	The role of the person responsible for the efficient and effective operation of the process, including process evaluation, quality management of inputs/outputs, initiating and managing improvement changes, securing resources, mentoring process participants.

1) Minimum Viable Role Model for the ministry domain management level

As a result of the study, the following Minimum Viable Role Model has been defined for the ministerial management level. The role model is unified and suitable for all ministries. The following table shows the roles and responsibilities.

TABLE V. MINIMUM VIABLE ROLE MODEL FOR THE MINISTRY DOMAIN MANAGEMENT LEVEL

Role	Responsibilities
Secretary General of the ministry	Gives mandate to team members. Measures performance. Authorizes budgets.
Business Architect of the ministry	Manages all components of the business architecture, including information architecture, service architecture, process architecture. Manages portfolios: services, processes, products, development needs.
Service Owner of the Agency	Responsible for managing the development needs of the agency and presenting them on ministerial portfolio management level.
ICT Architect of the ministry	Manages technology architecture, information systems architecture and applications architecture.
Data Architect of the ministry	Manages data architecture.

2) Minimum Viable Role Model for the agency management level

As a result of the study, the following Minimum Viable Role Model has been defined for the agency management level. The role model is unified and suitable for all agencies. The following table shows the roles and responsibilities.

TABLE VI. MINIMUM VIABLE ROLE MODEL FOR THE AGENCY MANAGEMENT LEVEL

Role	Responsibilities
Senior Executive	Authorises the service delivery budget, secures funding for product development.
Business Architect	Manages business architecture, including information architecture, service architecture, process architecture. Manages portfolios: services, processes, products, development needs.
ICT Architect	Manages technology architecture, information systems architecture and applications architecture.
Data Architect	Manages the organisation's data architecture.
Service Owner	Manages customer relationships, authorises service changes, defines service requirements, measures service performance.
Process Owner	Identifies processes, maps and models processes (as-is view), assesses process gaps and improvement opportunities, designs new processes (to-be view), coordinates implementation of changed processes, measures process performance, initiates and leads process changes.
Product Manager	Defines and communicates the product vision/strategy.
Product Owner	Ensures that the solution meets business requirements, maps product requirements, designs products to meet requirements, oversees product implementation, testing and delivery, defines and communicates product development needs (backlog), orders developments for the product, prioritises product developments, advises and trains product users.
Development Manager	Manages the project portfolio.
Project Manager	Leads product development project tasks, manages project scope, budget, schedule, resources, quality and risks.
IT Team Member	Performs minor development, maintains applications.

C. Additional findings

An unexpected positive result was found, which was not within the scope of the study's objectives.

Performance-based budgeting (officially called Activity-Based Budgeting in Estonia, also known as the TERE project), which was fully implemented in 2020, is an integral part of the state reform aimed at achieving more effective and efficient performance of public functions, higher quality of public services, reduction of general government expenditure and personnel costs, and more flexible and less bureaucratic management of the government sector [16] [17]. The initiative has had many critics, including the National Audit Office in 2023, which stated that: "activity-based budgeting, as it is referred to, remains largely regarded within the ministries as a project requested by the Ministry of Finance, with unclear objectives and benefits. In practice, real-world operations adhere to different formats." [18].

After introducing the semantic model to various ministries and agencies, several of the stakeholders responded that the proposed semantic model helps to align activity-based budgeting with real-world practices while managing services, process activities and consumed resources. Thus, this semantic model not only helps to govern and manage technology, but also benefits government performance management by being understandable to high-level government decision makers.

VI. CONCLUSION

A. Summary of findings

This study underlines that achieving interoperability in e-government requires top-level management mandates to drive change and effectively measure performance. Clear and formally agreed semantics of managed objects are essential to promote mutual understanding between stakeholders. In addition, defined roles and responsibilities must be operationalised and supported by the appropriate capabilities to ensure their successful implementation. These measures are essential to mitigate resource waste, reduce complexity and minimise risk in technology governance.

B. Recommendations

As the success criteria outlined in this study are applicable to both the public and private sectors, the proposed model is unified and reusable. Organisations and businesses can use this model to assess their current practices and make the necessary adjustments to improve their operational efficiency and collaboration.

C. Future research directions

- Evaluate the practical implications of the proposed models for improving collaboration and efficiency across government agencies.
- Conduct a longitudinal study to monitor Estonia's progress in implementing these frameworks.
- Investigate the applicability of the proposed model in other countries, as similar challenges are observed globally.

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